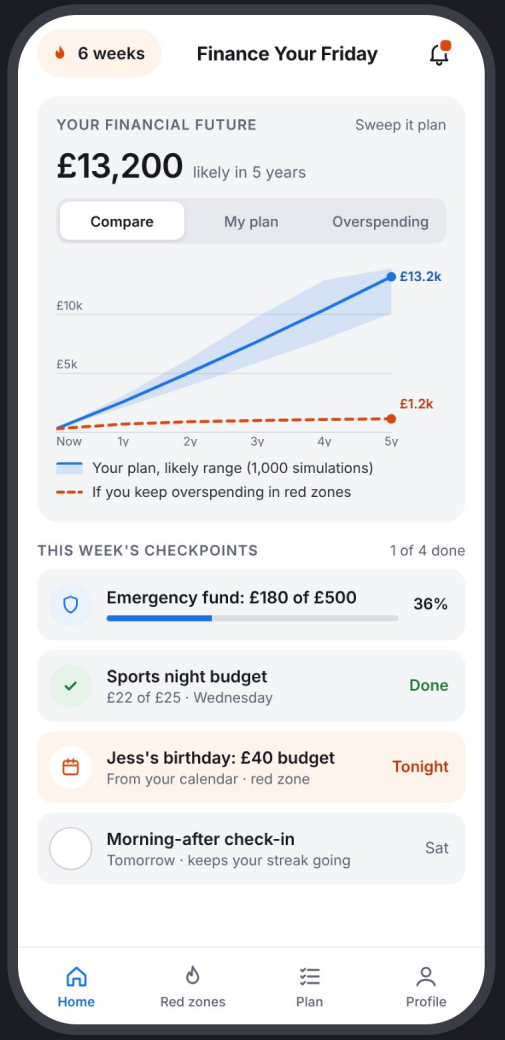
GreatUniHack 2026 · NatWest challenge

Finance Your Friday

An app that predicts when you are likely to overspend, and helps you plan for it.

Goldman Stanley - Sai Krishang Kasam, Arpit Bhabhra, Ching Hey Lau, An Han



Who it's for

Students and working families with no safety net

23%

of UK adults could not pay a surprise £850 bill. For parents, it is 33%.

£502

is the average monthly gap between a student's loan and their living costs.

13.1m

UK adults have low financial resilience. One in ten has no cash savings.

For these people, one big night or one surprise bill can mean an overdraft. They need a warning before it happens, and help the day after.

Sources: ONS Opinions and Lifestyle Survey, March 2025 · Save the Student, 2025 · FCA Financial Lives, 2024

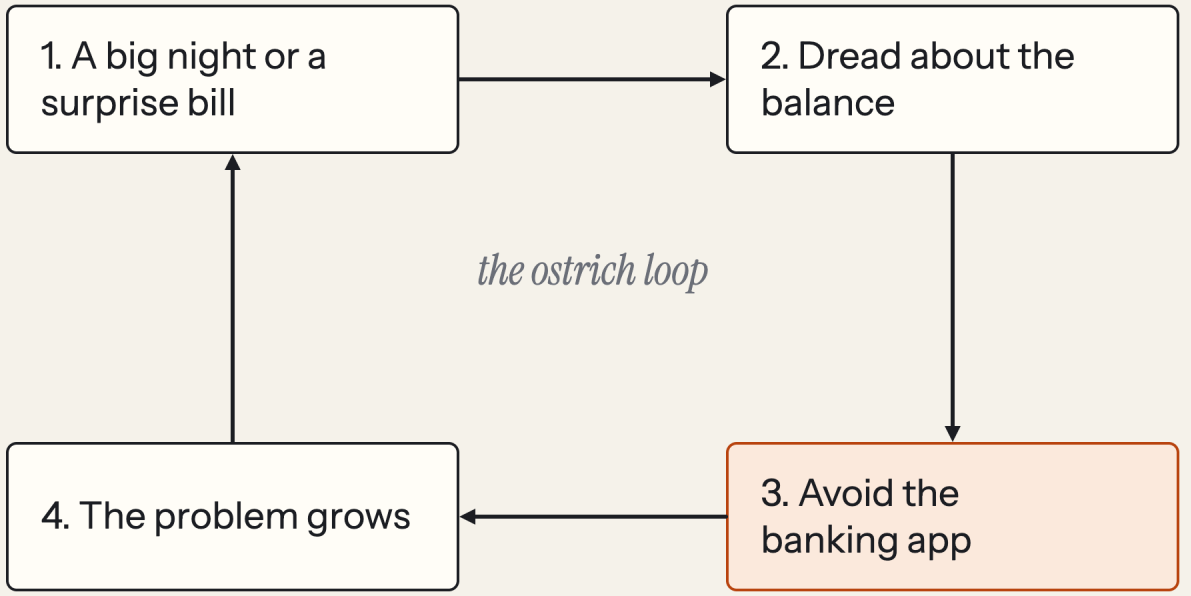
The ostrich effect

When people expect bad news, they stop looking

22% of UK adults avoid their bank balance because of money worries.

43% of 18 to 27-year-olds do the same.

50–80% fewer account checks by investors when markets fell.



Our morning-after check-in breaks the loop at step 3.

Sources: Aldermore and Opinium, 3,000 UK adults, 2025 · Karlsson, Loewenstein and Seppi, Journal of Risk and Uncertainty, 2009

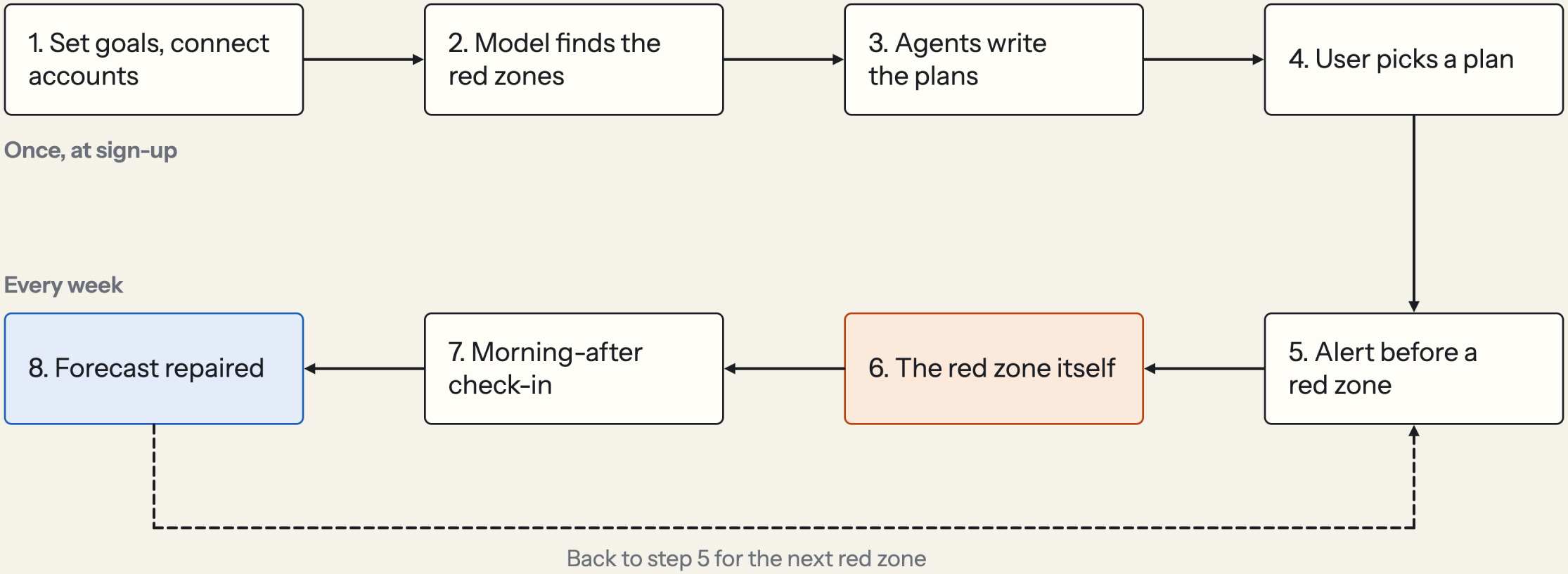
Our idea

Most overspending happens at moments you can predict. We warn people before those moments, and help them recover after.

We call these moments red zones: payday weekends, birthdays, work parties, Christmas.

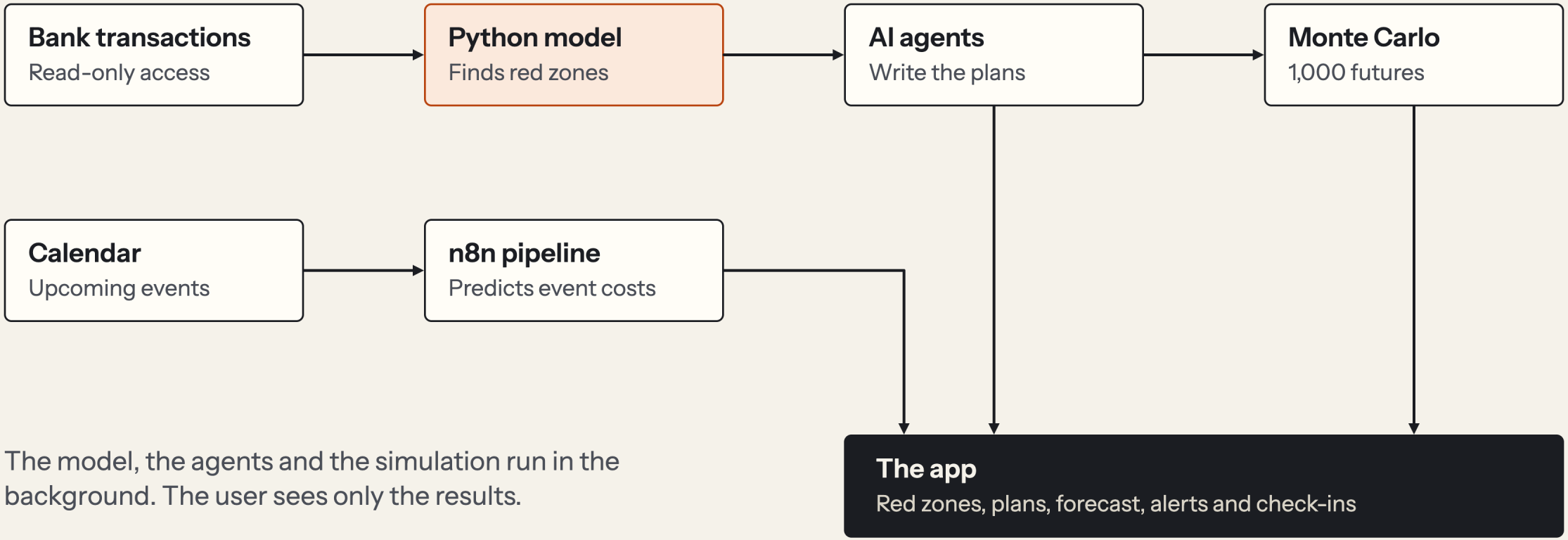
The user journey

Predict, plan, then recover every week



How it works

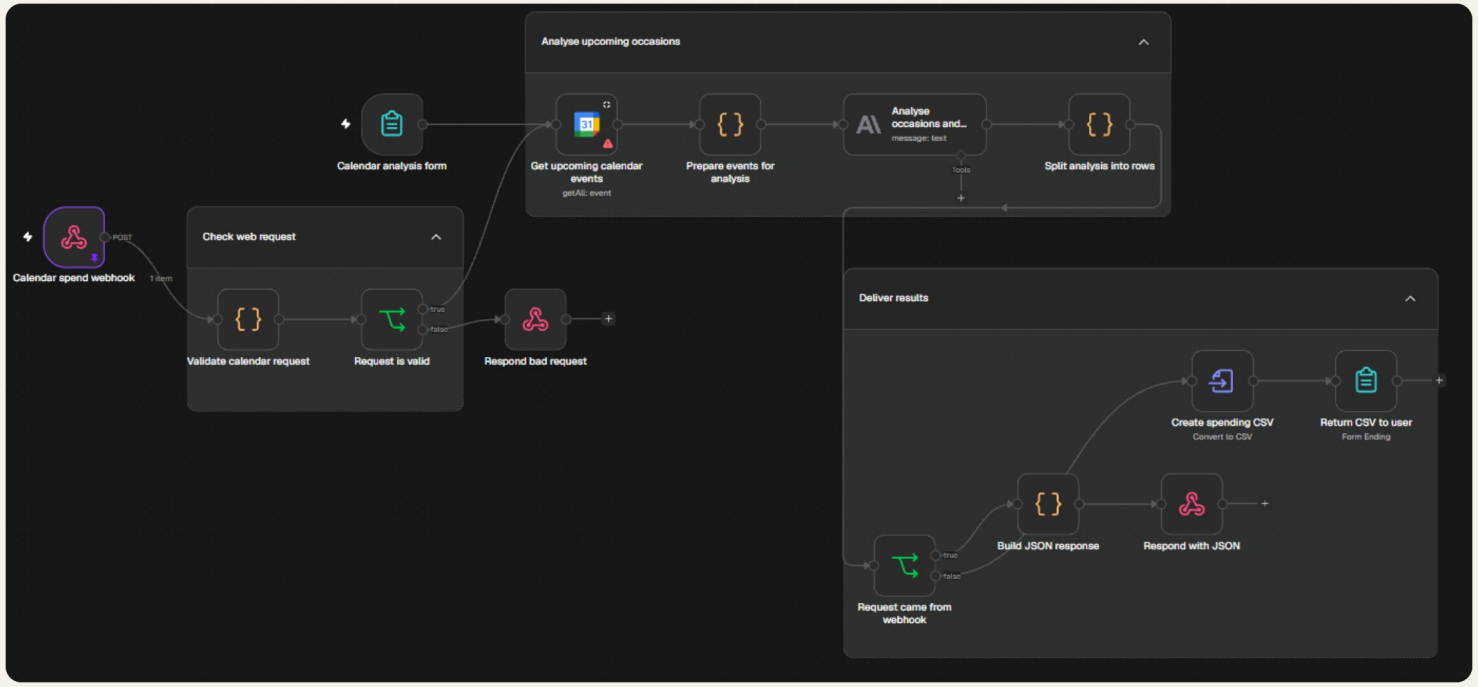
From bank data and a calendar to a plan



The model, the agents and the simulation run in the background. The user sees only the results.

The calendar pipeline

How n8n predicts what an event will cost

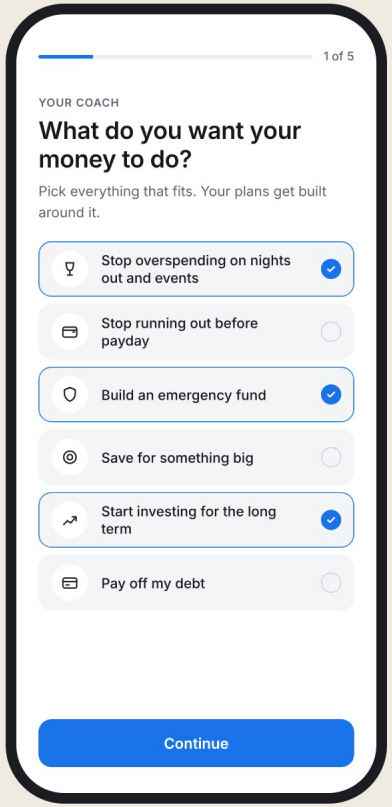


1. The app calls a webhook. A form also works, for testing.
2. n8n checks the request, then fetches upcoming calendar events.
3. Claude analyses each occasion and predicts what it will cost.
4. n8n sends the results to the app as JSON, or to the form as a CSV.

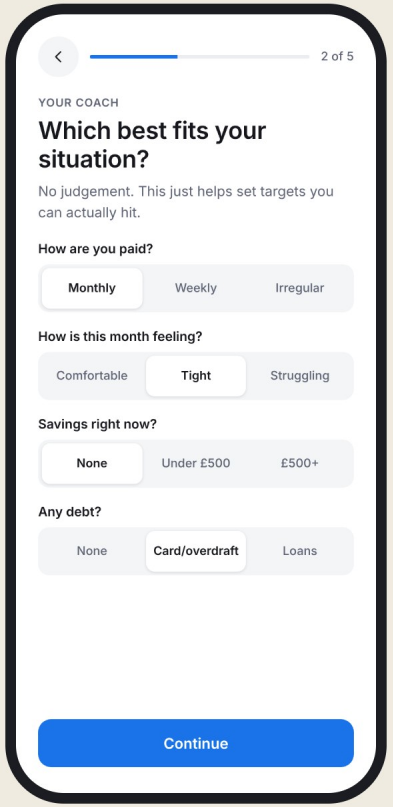
Demo · Sign-up

The user tells the coach what they want

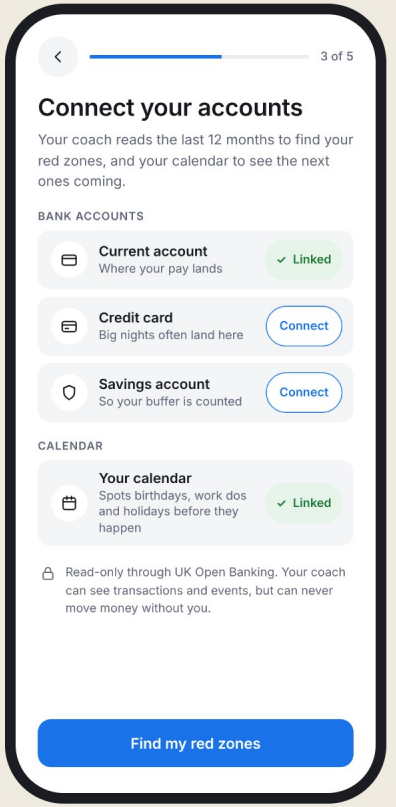
- Choose goals, such as fewer big nights out or an emergency fund
- Describe their situation in four quick questions
- Connect bank accounts and a calendar, read-only



Goals



Situation

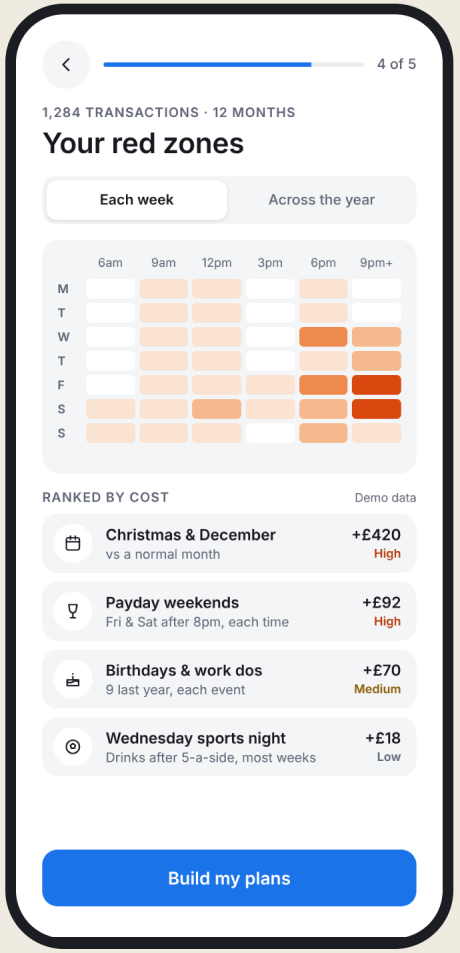


Accounts and calendar

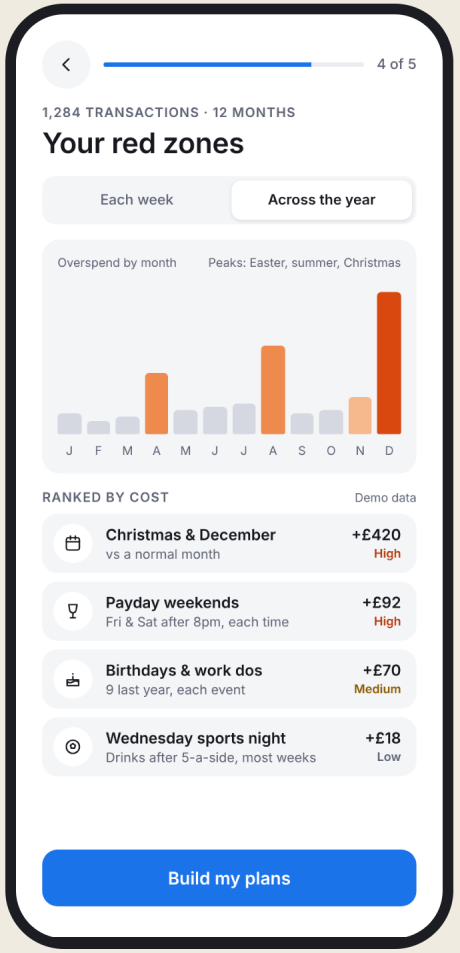
Demo · Red zones

Maya's red zones, ranked by cost

- The model reads 12 months of transactions and flags unusual spending
- Each week, her red zones are payday weekends and sports nights
- Across the year, they are Easter, the summer and Christmas
- Each red zone shows how much more she spends than usual



Each week

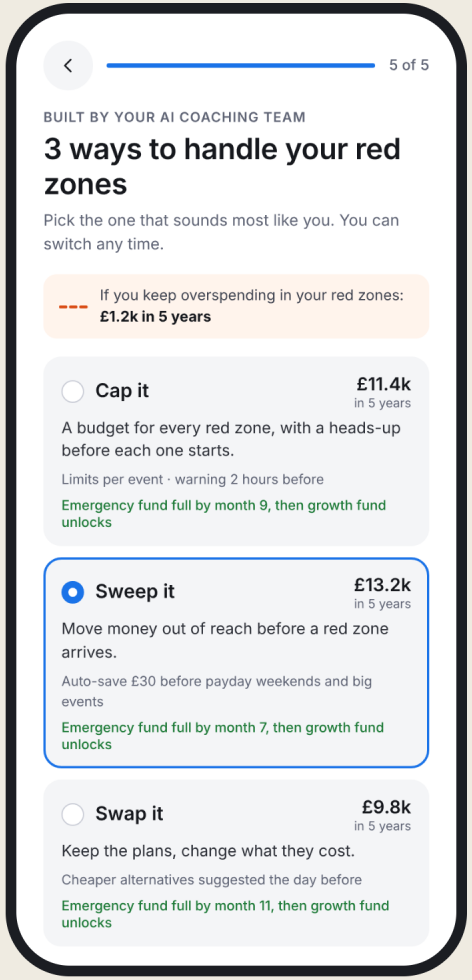


Across the year

Demo · Plans

Plans written for one person

- The agents read that person's red zones, income and goals
- They write several plans for those red zones, so no two users get the same plans
- Each plan shows its forecast next to the forecast if nothing changes
- The user picks the plan that suits them

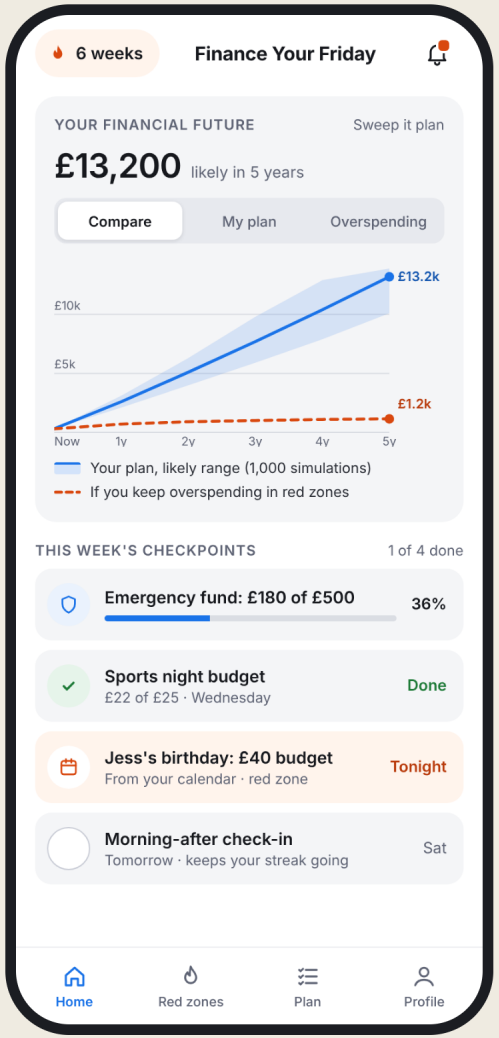


Example: the plans written for Maya

Demo · Home

Her future at the top, this week below

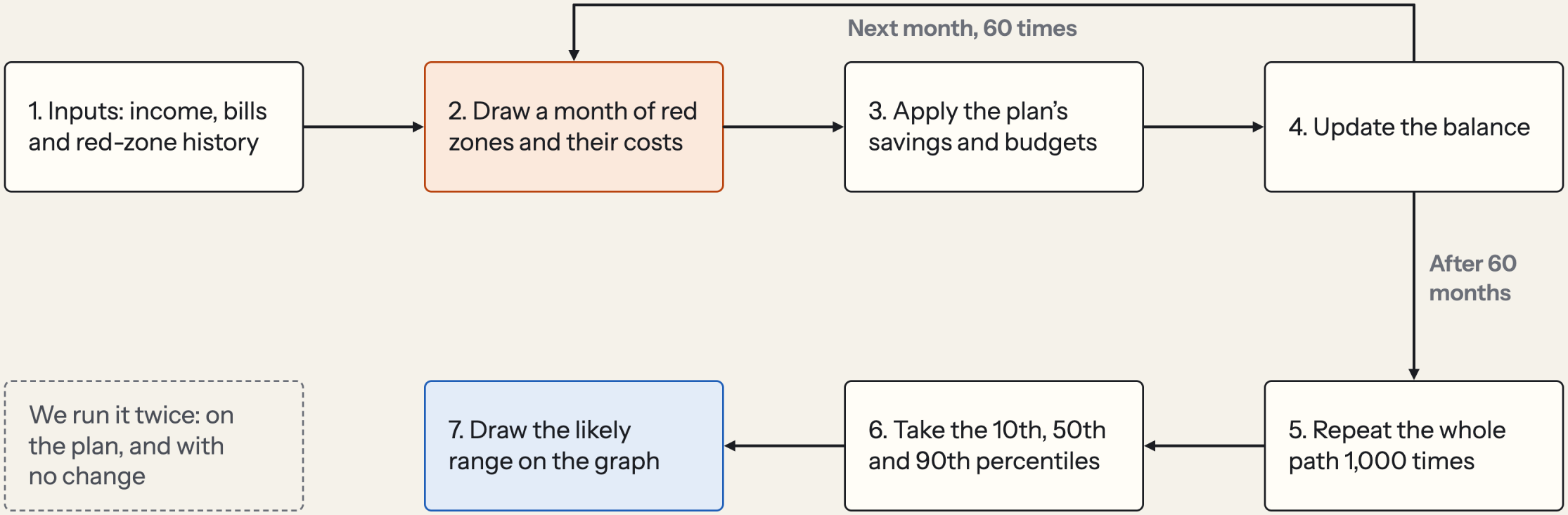
- The graph shows 1,000 simulated futures on her plan
- The dashed line shows what happens if she keeps overspending in her red zones
- Checkpoints turn the plan into steps for this week
- Red zones from her calendar appear before they happen



The Monte Carlo forecast

How we simulate 1,000 futures

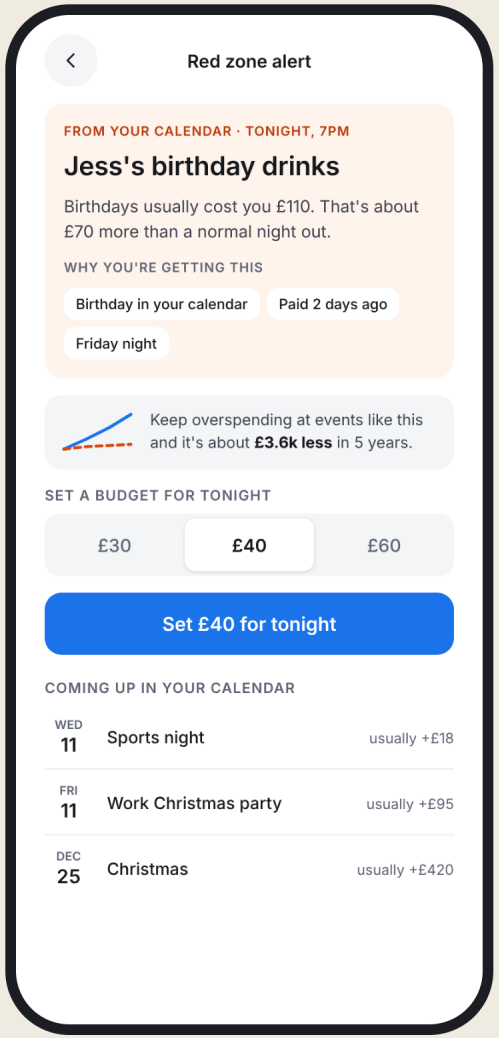
Each run is one possible future for that person. Together, the runs give the likely range.



Demo · Red zone alert

A warning before it happens

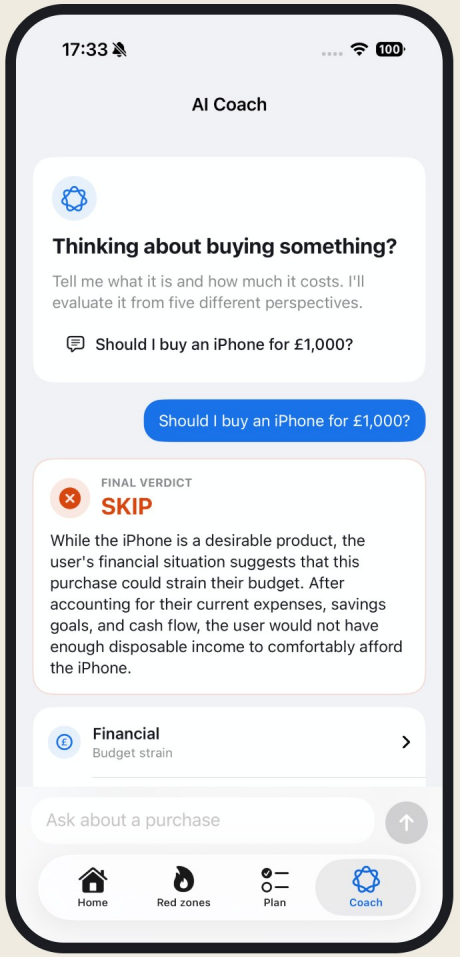
- The n8n pipeline finds a birthday in her calendar and tags it
- The model predicts what the night will cost her, and says why
- She sets a budget for tonight with one tap
- She can also see her next red zones



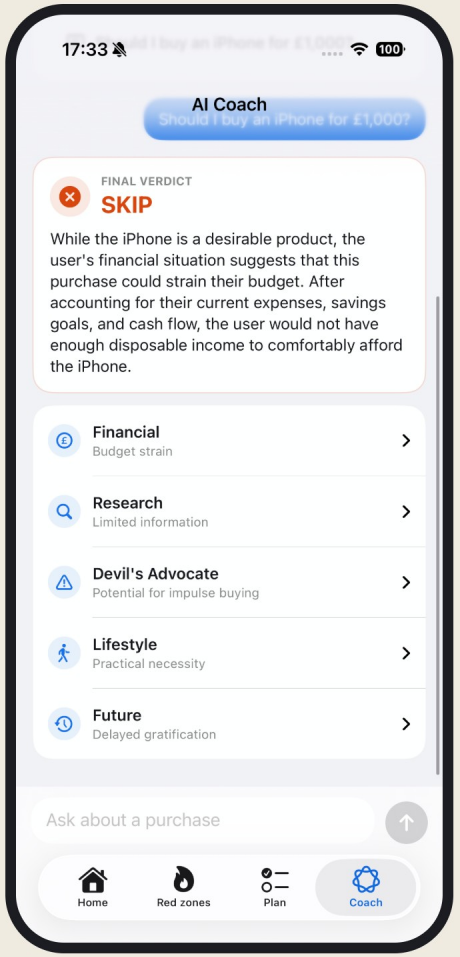
Demo · AI coach

Ask before a big purchase

- Maya asks in plain words: “Should I buy an iPhone for £1,000?”
- Five agents review it: financial, research, devil’s advocate, lifestyle and future
- The coach gives one verdict and the reason for it
- She can open each agent’s view with one tap



The verdict

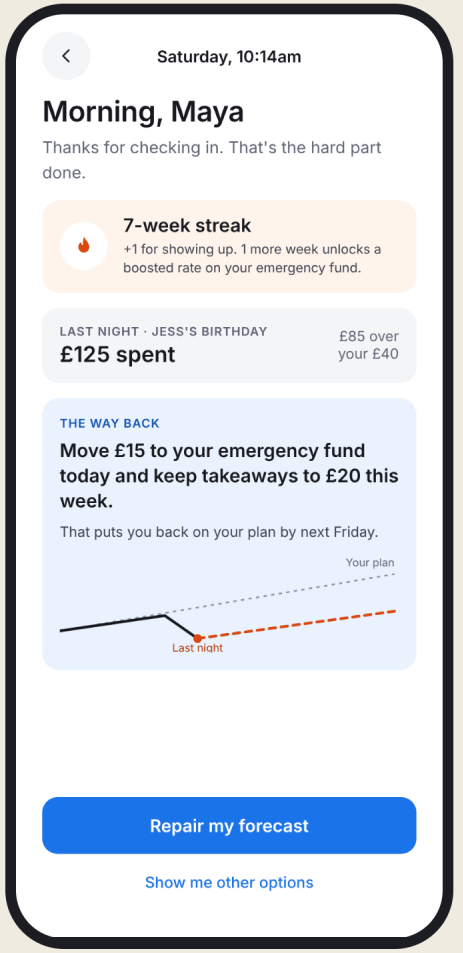


Five agent views

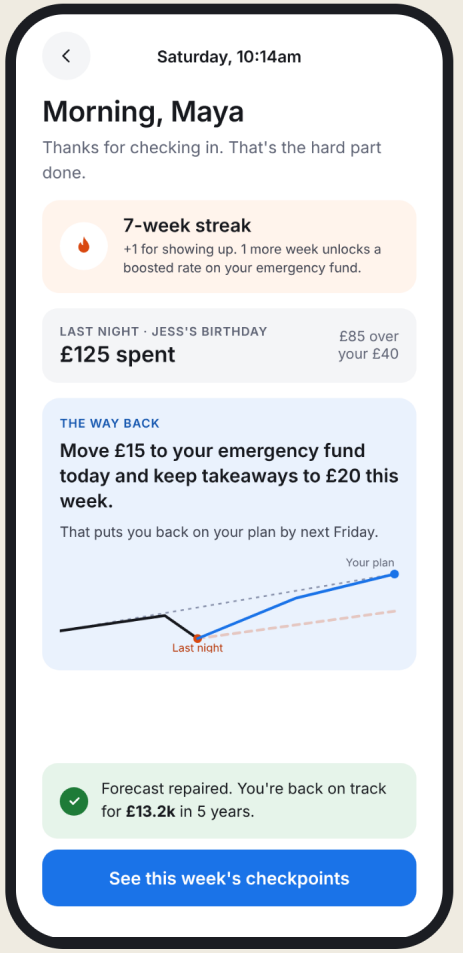
Demo · The morning after

Opening the app earns a reward

- Her streak goes up because she checked in, even after a bad night
- She sees one number, then one fix
- One tap repairs her forecast
- A longer streak unlocks a better savings rate from the bank



Check-in



Forecast repaired

The NatWest brief

How we meet each criterion

Criterion	What we built
Customer focus	Built for people with no safety net, at the moments they overspend
Innovation	Predicts when someone is likely to overspend, from spending and calendar data
Social impact	Aimed at students and families, who are hit hardest by surprise costs
AI and data	Anomaly model, n8n pipeline, AI agents for plans and purchases, and Monte Carlo forecasts
Feasibility	Uses Open Banking data, and a streak reward a bank can offer now

Finance Your Friday

A warning before the red zone. Help the morning after.

Thank you. Any questions?

Goldman Stanley · GreatUniHack 2026